

911 Calls Data Analysis

A data analysis project exploring emergency 911 call data from Montgomery County, Pennsylvania. This project utilizes Python and key data science libraries to perform exploratory data analysis (EDA), uncover trends, and generate actionable insights.

Project Overview

This project analyzes a real-world dataset of 911 emergency calls to answer important questions about when and why people call 911. It focuses on discovering patterns in call volumes, call types, and their geographical and temporal distributions.

Data Set - <https://www.kaggle.com/datasets/mchirico/montcoalert>

Objectives

- Understand the structure and content of the 911 call dataset.
- Analyze the frequency of different types of emergencies.
- Explore temporal patterns (hour, day, month).
- Visualize trends across different townships.
- Extract useful insights for emergency response management.

Tools & Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Dataset

- **Source:** Kaggle - 911 Calls
- **Size:** ~1.5 million rows
- **Fields:**

- lat, lng – Location coordinates
- desc – Description of the emergency
- zip, title, timeStamp, twp, addr, e – Other metadata

Key Insights

- The most common reason for 911 calls was **EMS-related** emergencies.
- Call volumes peak during **evening hours** and vary by **day of the week**.
- Certain **townships** have significantly higher call volumes.
- There are **seasonal variations** in call trends.

Visualizations

The analysis includes:

- Count plots of emergency types
- Time series line graphs for call trends
- Heatmaps of calls by hour and day
- Geographic visualizations of call distributions

Learnings

- Gained hands-on experience in handling time-series and categorical data.
- Improved data visualization skills using Seaborn and Matplotlib.
- Practiced feature engineering (e.g., extracting hour, month from timestamp).
- Learned to derive actionable insights from real-world emergency data.

Future Improvements

- Integrate **Geopandas** for advanced mapping.
 - Apply **clustering algorithms** to detect emergency hotspots.
 - Create an **interactive dashboard** using **Plotly** or **Power BI**.
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